



COURSE CODE/TITLE:	CPE212 Automating Server Management	SECTION:	CPE31S2
DATE PERFORMED:	08/27/2026	DATE SUBMITTED:	08/27/2026
NAME:	Melquisedec D. Iquin	INSTRUCTOR:	Engr. Jimlord Quejado

ACTIVITY NO. 5.1 **Implement Ansible roles in playbooks**

1. PROCEDURE OUTPUT

What procedures did you perform?

```
iquin@workstation: ~/CPE212_IQUIN
Receiving objects: 100% (9/9), done.
remote: Total 9 (delta 0), reused 3 (delta 0), pack-reused 0 (from 0)
iquin@workstation:~$ cd CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ls
install_apache.yml  README.md
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.125]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: iquin@192.168.56.125: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
ok: [192.168.56.124]
ok: [192.168.56.123]

TASK [update repository index] *****

changed: [192.168.56.123]
changed: [192.168.56.124]

TASK [install apache2 package] *****

changed: [192.168.56.123]
changed: [192.168.56.124]

TASK [add PHP support for apache] *****

changed: [192.168.56.123]
changed: [192.168.56.124]

PLAY RECAP *****
```

```
GNU nano 7.2                                install_apache.yml
---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        when: ansible_distribution == "Ubuntu"
```

- This is the outdated version compared to the ones we later did, comparing them functions the same way, but as for the later note its way shorter and more flexible.

```
iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ nano install_apache.yml
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook --ask-become-pass install_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.125]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: iquin@192.168.56.125: Permission denied (publickey,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
ok: [192.168.56.124]
ok: [192.168.56.123]

TASK [update repository index] *****
changed: [192.168.56.123]
changed: [192.168.56.124]

TASK [install apache2 package] *****
ok: [192.168.56.124]
ok: [192.168.56.123]

TASK [add PHP support for apache] *****
ok: [192.168.56.124]
ok: [192.168.56.123]

PLAY RECAP *****
192.168.56.123      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.124      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.125      : ok=0    changed=0    unreachable=1    failed=0    s
```

```
---
- hosts: all
  become: true
  tasks:

    - name: update repository index
      apt:
        update_cache: yes
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      apt:
        name: apache2
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: add PHP support for apache
      apt:
        name: libapache2-mod-php
        state: latest
      when: ansible_distribution == "Ubuntu"

    - name: install apache2 package
      dnf:
        name: httpd
        state: latest
      when: ansible_distribution == "CentOS"
```

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2                                install_apache.yml *
```

```
- name: install apache2 package
  apt:
    name: apache2
    state: latest
    when: ansible_distribution == "Ubuntu"

- name: add PHP support for apache
  apt:
    name: libapache2-mod-php
    state: latest
    when: ansible_distribution == "Ubuntu"

- name: update repository index
  dnf:
    update_cache: yes
    when: ansible_distribution == "CentOS"

- name: install apache2 package
  dnf:
    name: httpd
    state: latest
    when: ansible_distribution == "CentOS"

- name: add PHP support for apache
  dnf:
    name: php
    state: latest
    when: ansible_distribution == "CentOS"
```

- Well the result still functions the same as before, it installed everything along with apache2 packages etc.

```

iquin@workstation: ~/CPE212_IQUIN
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --ask-become-pass in
stall_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.124]
ok: [192.168.56.125]

TASK [update repository index] *****
skipping: [192.168.56.125]
changed: [192.168.56.124]
changed: [192.168.56.123]

TASK [install apache2 package] *****
skipping: [192.168.56.125]
ok: [192.168.56.124]
ok: [192.168.56.123]

TASK [add PHP support for apache] *****
skipping: [192.168.56.125]
ok: [192.168.56.124]

TASK [install apache2 package] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

TASK [add PHP support for apache] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

PLAY RECAP *****
192.168.56.123      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.124      : ok=4    changed=1    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0
192.168.56.125      : ok=4    changed=0    unreachable=0    failed=0    s
kipped=3    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$

```

TASK 2:

```
iquin@workstation: ~/CPE212_IQUIN
GNU nano 7.2 install_apache.yml *
--
- hosts: all
  become: true
  tasks:
    - name: update repository index Ubuntu
      apt:
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: install apache2 package and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        when: ansible_distribution == "Ubuntu"

    - name: update repository index for CentOS
      dnf:
        update_cache: yes
        when: ansible_distribution == "CentOS"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php

^G Help      ^O Write Out ^W Where Is  ^K Cut       ^T Execute   ^C Location
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line
```

```
GNU nano 7.2                                install_apache.yml *
```

```
tasks:
  - name: update repository index Ubuntu
    apt:
      update_cache: yes
      when: ansible_distribution == "Ubuntu"

  - name: install apache2 package and php packages for Ubuntu
    apt:
      name:
        - apache2
        - libapache2-mod-php
      state: latest
      when: ansible_distribution == "Ubuntu"

  - name: update repository index for CentOS
    dnf:
      update_cache: yes
      when: ansible_distribution == "CentOS"

  - name: install apache and php packages for CentOS
    dnf:
      name:
        - httpd
        - php
      state: latest
      when: ansible_distribution == "CentOS"
```

```
iquin@workstation: ~/CPE212_IQUIN
```

```
iquin@workstation:~/CPE212_IQUIN$ nano install_apache.yml
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --ask-become-pass in
stall_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.124]
ok: [192.168.56.125]

TASK [update repository index] *****
skipping: [192.168.56.125]
changed: [192.168.56.124]
changed: [192.168.56.123]

TASK [install apache2 package and php packages for Ubuntu] *****
skipping: [192.168.56.125]
ok: [192.168.56.123]
ok: [192.168.56.124]

TASK [update repository index for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

TASK [install apache and php packages for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]
```

```
iquin@workstation: ~/CPE212_IQUIN

TASK [update repository index] *****
skipping: [192.168.56.125]
changed: [192.168.56.124]
changed: [192.168.56.123]

TASK [install apache2 package and php packages for Ubuntu] *****
skipping: [192.168.56.125]
ok: [192.168.56.123]
ok: [192.168.56.124]

TASK [update repository index for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

TASK [install apache and php packages for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

PLAY RECAP *****
192.168.56.123      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.124      : ok=3    changed=1    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.125      : ok=3    changed=0    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
```

```
GNU nano 7.2      install_apache.yml
---
- hosts: all
  become: true
  tasks:

    - name: install apache2 package and php packages for Ubuntu
      apt:
        name:
          - apache2
          - libapache2-mod-php
        state: latest
        update_cache: yes
        when: ansible_distribution == "Ubuntu"

    - name: update repository index for CentOS
      dnf:
        update_cache: yes
        when: ansible_distribution == "CentOS"

    - name: install apache and php packages for CentOS
      dnf:
        name:
          - httpd
          - php
        state: latest
        update_cache: yes
        when: ansible_distribution == "CentOS"
```

```
iquin@workstation: ~/CPE212_IQUIN
PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.124]
ok: [192.168.56.123]
ok: [192.168.56.125]

TASK [install apache2 package and php packages for Ubuntu] *****
skipping: [192.168.56.125]
ok: [192.168.56.124]
ok: [192.168.56.123]

TASK [update repository index for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

TASK [install apache and php packages for CentOS] *****
skipping: [192.168.56.123]
skipping: [192.168.56.124]
ok: [192.168.56.125]

PLAY RECAP *****
192.168.56.123      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.124      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=2    rescued=0    ignored=0
192.168.56.125      : ok=3    changed=0    unreachable=0    failed=0    s
kipped=1    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$
```

```
GNU nano 7.2 install_apache.yml
--
- hosts: all
  become: true
  tasks:

    - name: install apache and php
      apt:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

```
iquin@workstation: ~/CPE212_IQUIN

TASK [install apache and php] *****
fatal: [192.168.56.123]: FAILED! => {"msg": "The task includes an option with an
undefined variable. The error was: 'apache_package' is undefined. 'apache_packa
ge' is undefined\n\nThe error appears to be in '/home/iquin/CPE212_IQUIN/install
_apache.yml': line 6, column 7, but may\nbe elsewhere in the file depending on t
he exact syntax problem.\n\nThe offending line appears to be:\n\n    - name: i
ninstall apache and php\n    ^ here\n"}
fatal: [192.168.56.124]: FAILED! => {"msg": "The task includes an option with an
undefined variable. The error was: 'apache_package' is undefined. 'apache_packa
ge' is undefined\n\nThe error appears to be in '/home/iquin/CPE212_IQUIN/install
_apache.yml': line 6, column 7, but may\nbe elsewhere in the file depending on t
he exact syntax problem.\n\nThe offending line appears to be:\n\n    - name: i
ninstall apache and php\n    ^ here\n"}
fatal: [192.168.56.125]: FAILED! => {"msg": "The task includes an option with an
undefined variable. The error was: 'apache_package' is undefined. 'apache_packa
ge' is undefined\n\nThe error appears to be in '/home/iquin/CPE212_IQUIN/install
_apache.yml': line 6, column 7, but may\nbe elsewhere in the file depending on t
he exact syntax problem.\n\nThe offending line appears to be:\n\n    - name: i
ninstall apache and php\n    ^ here\n"}

PLAY RECAP *****
192.168.56.123      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
192.168.56.124      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0
192.168.56.125      : ok=1    changed=0    unreachable=0    failed=1    s
kipped=0    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$ nano install_apache.yml
iquin@workstation:~/CPE212_IQUIN$
```

```
GNU nano 7.2                                install_apache.yml *
```

```
---
- hosts: all
  become: true
  tasks:

    - name: install apache and php
      package:
        name:
          - "{{ apache_package }}"
          - "{{ php_package }}"
        state: latest
        update_cache: yes
```

```
GNU nano 7.2                                hosts *
```

```
[myvirtualmachines]
192.168.56.123 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.124 apache_package=apache2 php_package=libapache2-mod-php
192.168.56.125 ansible_user=melquisedec apache_package=httpd php_package=php
```

```
iquin@workstation:~/CPE212_IQUIN$ nano install_apache.yml
iquin@workstation:~/CPE212_IQUIN$ nano install_apache.yml
iquin@workstation:~/CPE212_IQUIN$ nano hosts
iquin@workstation:~/CPE212_IQUIN$ ansible-playbook -i hosts --ask-become-pass in
stall_apache.yml
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.124]
ok: [192.168.56.125]

TASK [install apache and php] *****
ok: [192.168.56.124]
ok: [192.168.56.123]
ok: [192.168.56.125]

PLAY RECAP *****
192.168.56.123      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.124      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.125      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0

iquin@workstation:~/CPE212_IQUIN$
```

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

```

BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.125]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: melquisedec@192.168.56.125: Permission denied (public key,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
fatal: [192.168.56.124]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: melquisedec@192.168.56.124: Permission denied (public key,password).", "unreachable": true}
fatal: [192.168.56.123]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: melquisedec@192.168.56.123: Permission denied (public key,password).", "unreachable": true}

PLAY RECAP *****
192.168.56.123      : ok=0    changed=0    unreachable=1    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.124      : ok=0    changed=0    unreachable=1    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.125      : ok=0    changed=0    unreachable=1    failed=0    s
kipped=0    rescued=0    ignored=0

[melquisedec@localhost CPE212_IQUIN]$

```

```

BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
fatal: [192.168.56.125]: UNREACHABLE! => {"changed": false, "msg": "Failed to connect to the host via ssh: melquisedec@192.168.56.125: Permission denied (public key,gssapi-keyex,gssapi-with-mic,password).", "unreachable": true}
ok: [192.168.56.123]
ok: [192.168.56.124]

TASK [install apache and php] *****
ok: [192.168.56.123]
ok: [192.168.56.124]

PLAY RECAP *****
192.168.56.123      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.124      : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0    rescued=0    ignored=0
192.168.56.125      : ok=0    changed=0    unreachable=1    failed=0    s
kipped=0    rescued=0    ignored=0

[melquisedec@localhost CPE212_IQUIN]$

```

```
ml -K
BECOME password:

PLAY [all] *****

TASK [Gathering Facts] *****
ok: [192.168.56.123]
ok: [192.168.56.125]
ok: [192.168.56.124]

TASK [install apache and php] *****
ok: [192.168.56.124]
ok: [192.168.56.123]
ok: [192.168.56.125]

PLAY RECAP *****
192.168.56.123 : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0      rescued=0    ignored=0
192.168.56.124 : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0      rescued=0    ignored=0
192.168.56.125 : ok=2    changed=0    unreachable=0    failed=0    s
kipped=0      rescued=0    ignored=0

[melquisedec@localhost CPE212_IQUIN]$
```

Reflections:

Answer the following:

1. Why do you think refactoring of playbook codes is important?
 - Refactoring makes playbooks shorter, easier to read, and less the error by removing duplicate code, like the httpd typo you would've had to fix twice if the code stayed split.
2. When do we use the "when" command in playbook?
 - We use when to run a task only under certain conditions, such as targeting a task at a specific OS distribution like ubuntu or centos just like what I did earlier in the ubuntu workstation it basically lets one playbook manage multiple types of system like ubuntu and centos.

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used:

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.
claude

Purpose of Use:

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

- Figuring out how to connect my github repo to ubuntu workstation, along with grammatical error correcting

Extent of Use:

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.



**COMPUTER
ENGINEERING**

QUEZON CITY

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

Lab activities, templates, and related instructional materials shall not be reuploaded, reproduced, distributed, or shared with external organizations or third parties without the prior express written permission of the Technological Institute of the Philippines.